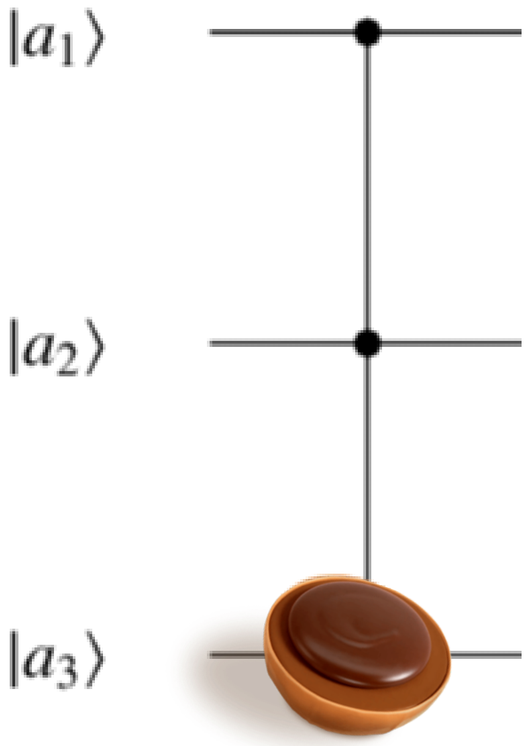
Alexandru Ioan Cuza University Iași,
Faculty of Computer Science

FreeYaMind
Campus

# Quantum Genome Detective Lab




# TOFFOLI

# TOFFIFEE

Cotea Carla, Enachescu Alexandra,
Pricop Tudor, Picioroagă Alexandru

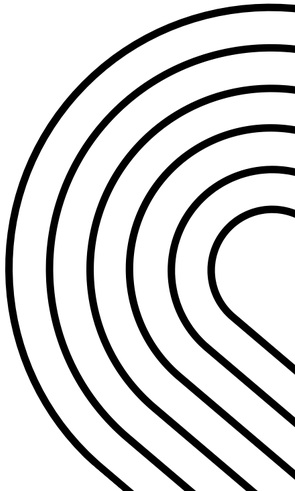

Evidence/
objective
QAOA search/
pruning
DIRECTIONS
Hardware/
Error Mitigation
Classic
Baseline/
Scaling

# Does evidence weighting improve reconstruction?

| METHOD | RULE |
|---|---|
| **A — Baseline** | Each observed SNP pair has weight 1. |
| **B1 — Per-read normalization** | Each informative read has total weight 1. |
| **B2 — Per-site normalization** | Each observed site has total incident pair weight 1. |
| **C — Deduplicated reads** | Remove exactly identical read rows, then apply baseline voting. |
| **D — Full-read MEC** | Minimize individual read corrections needed to fit either proposed copy. |

**Does changing the scoring rule improve reconstruction?**

Different weighting rules can change which reconstruction minimizes the objective.



# NO OBJECTIVE WINS ON EVERY TESTED INSTANCE

**EXACT COMPARISON**

12, 16, 20 and 30 SNPs

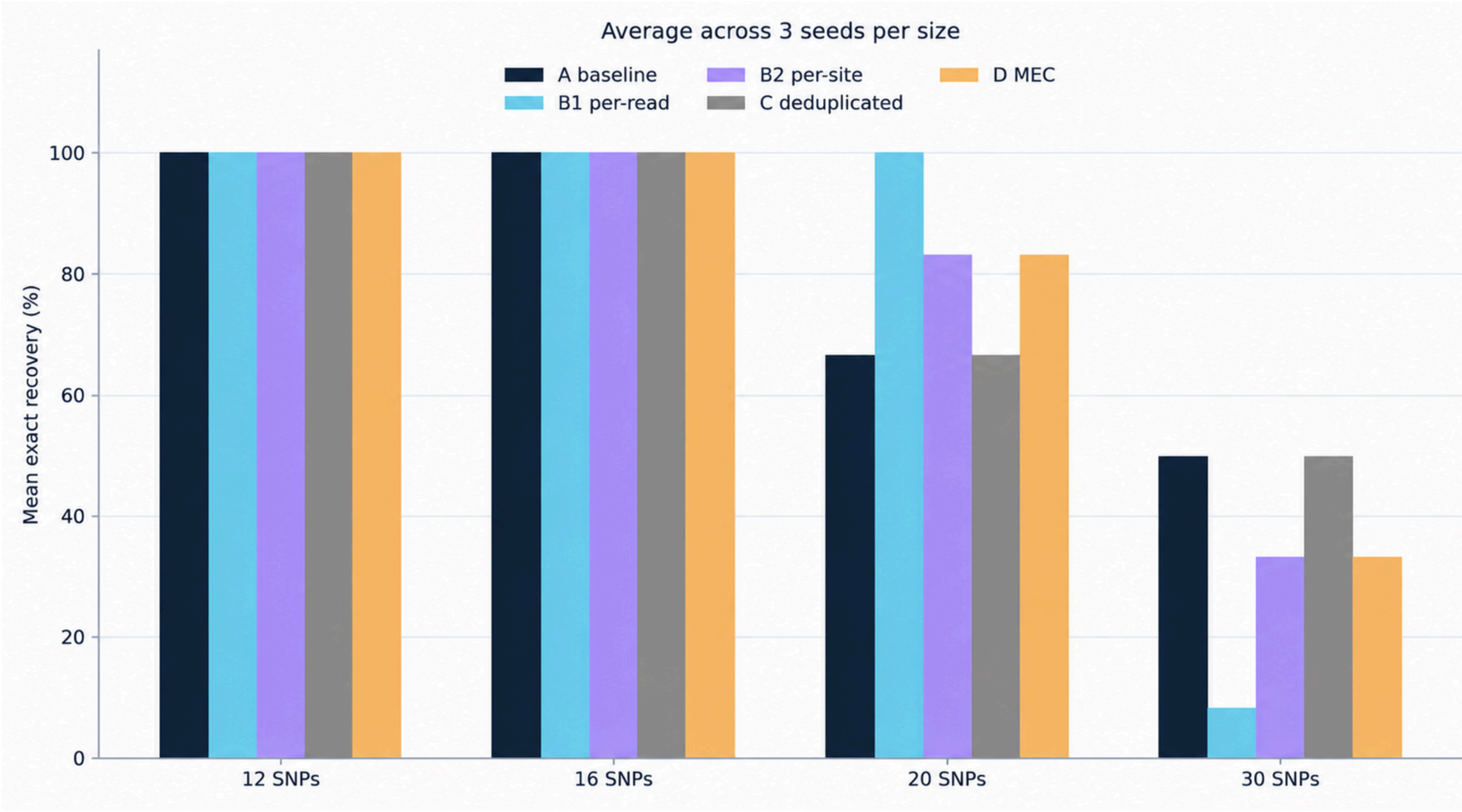
3 seeds per size

**OBSERVATION**

All methods recovered truth at 12–16 SNPs.

At 20 SNPs, B1 recovered truth on all three tested instances.

At 30 SNPs, B1 loses to baseline on two of three instances.




A lower model cost does not guarantee the true reconstruction.

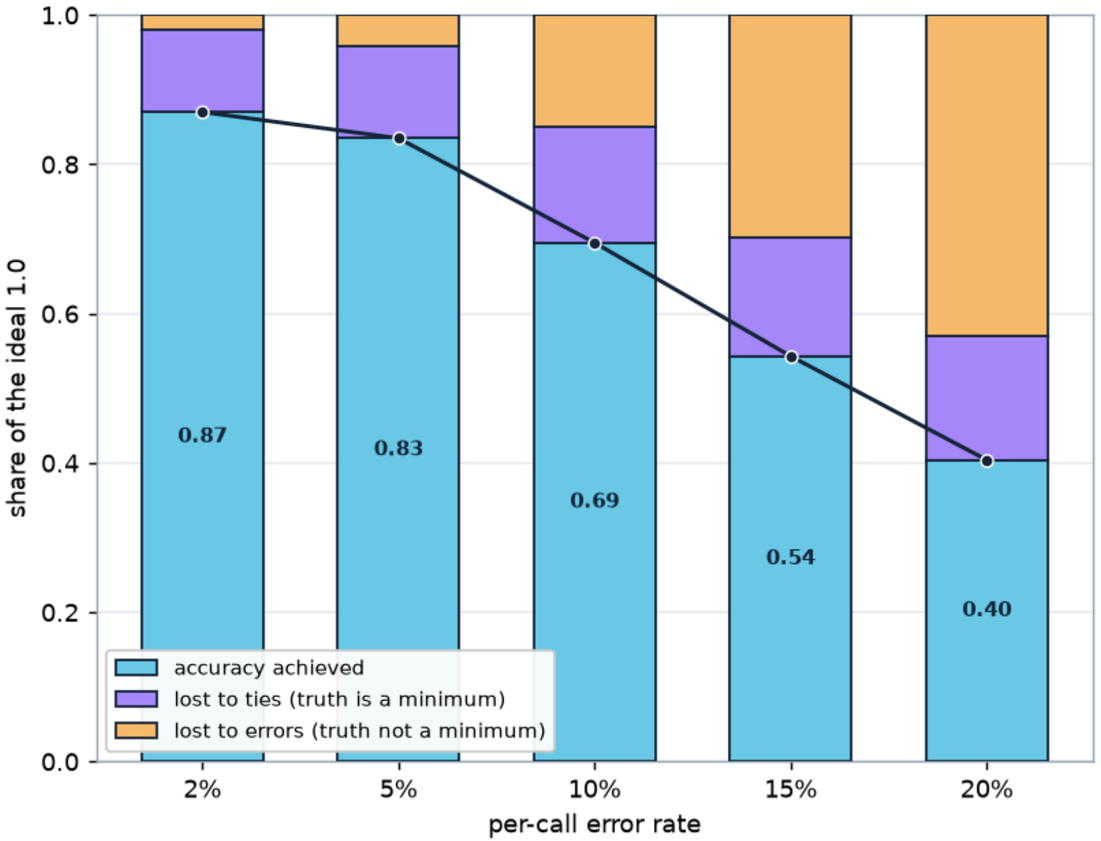
# Why is baseline accuracy ~0.696?  Errors AND ties - a decomposition

*No-pruning full model, n = 5 SNPs, seed-averaged (seeds 1-5). Accuracy = random_pick_accuracy (1/|tied set| when the truth is tied).*

## Decomposition by per-call error rate

| error | accuracy | truth is a minimum | mean tied set | accuracy when truth is min |
|---|---|---|---|---|
| 2% | 0.870 | 0.980 | 1.25 | 0.888 |
| 5% | 0.835 | 0.958 | 1.31 | 0.871 |
| 10% | 0.695 | 0.850 | 1.45 | 0.818 |
| 15% | 0.543 | 0.702 | 1.69 | 0.773 |
| 20% | 0.404 | 0.570 | 1.90 | 0.709 |

## Where the missing accuracy goes

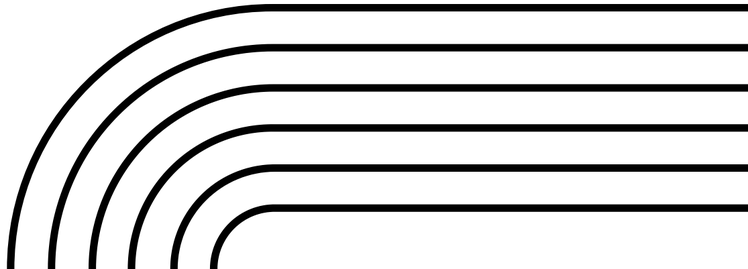



Errors dominate the trend (gold grows 0.02 -> 0.43 as error rises), but even a perfect optimizer cannot reach 1.0: the purple band is the honest tie penalty, and the tied-set grows with error too.

## Every pruning method tried - what it drops, and whether it helps

*Signed-graph haplotype phasing, n = 5 SNPs, seed-averaged (seeds 1-5). Green preserves connectivity; gold disconnects and harms; purple = control.*

| pruning rule | kind | drops edges where … | thresholds swept | fraction disconnected | accuracy | pruned min is full min | verdict |
|---|---|---|---|---|---|---|---|
| **full (none)** | baseline | - (keep every edge) | - | 0.24 | 0.696 | 1.00 | **baseline** |
| **net** | rule | \|A-D\| < t  (weak net) | t=0..3 | 0.64 | 0.547 | 0.52 | **harms** |
| **support** | rule | A+D < m  (few votes) | m=1..4 | 0.82 | 0.333 | 0.30 | **harms** |
| **tie** | rule | A == D  (no information) | - | 0.24 | 0.696 | 1.00 | **neutral** |
| **topk** | rule | keep only k strongest | k=0..10 | 0.78 | 0.488 | 0.49 | **harms** |
| **mstforest** | rule | max-weight spanning forest | - | 0.24 | 0.688 | 0.98 | **preserves** |
| **bridge** | rule | removal splits a component | - | 0.64 | 0.547 | 0.52 | **harms** |
| **random** | control | drop same count at random | - | 0.70 | 0.528 | 0.40 | **control** |
| **strong** | control | drop the strongest edges | - | 0.80 | 0.344 | 0.28 | **control** |

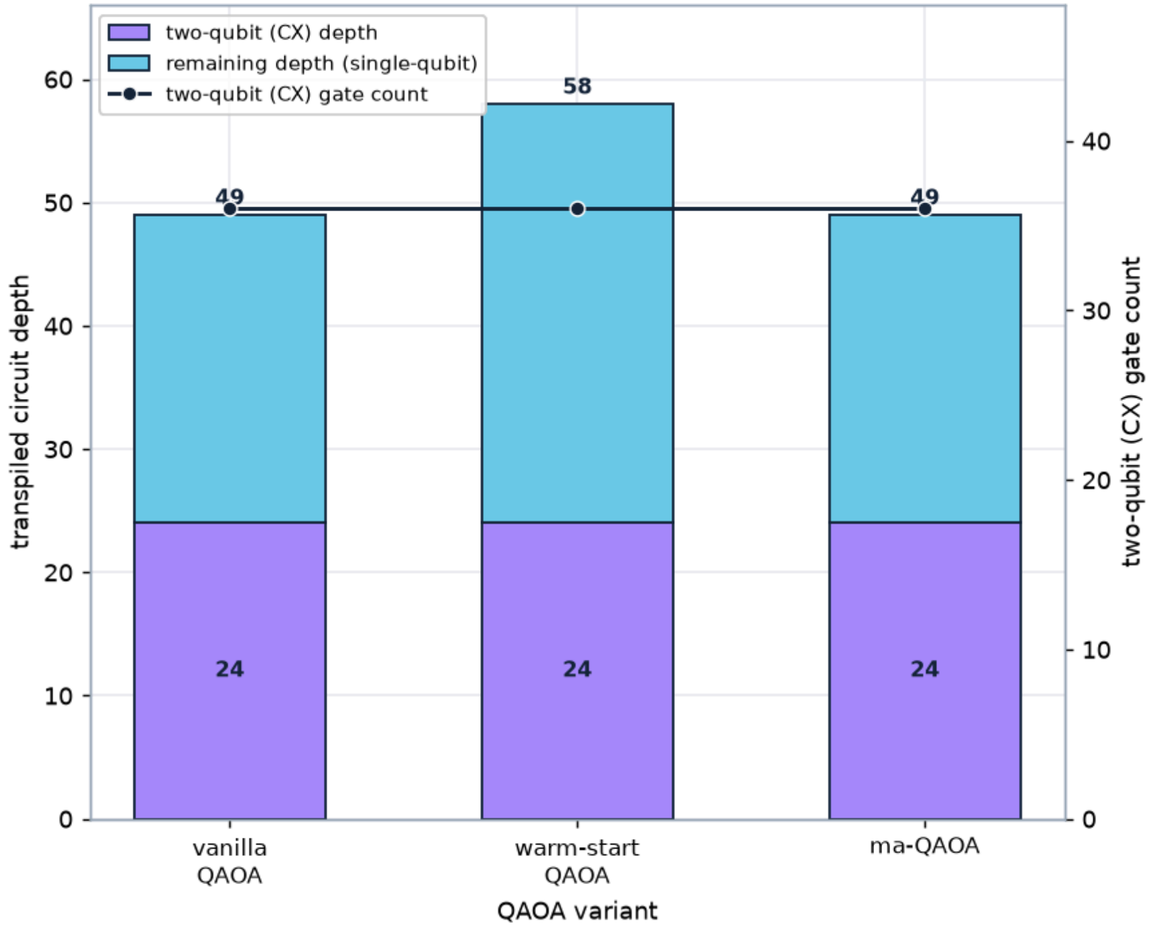


## Three QAOA variants, same instance - how do the circuit resources compare?

### Resource cost by QAOA variant

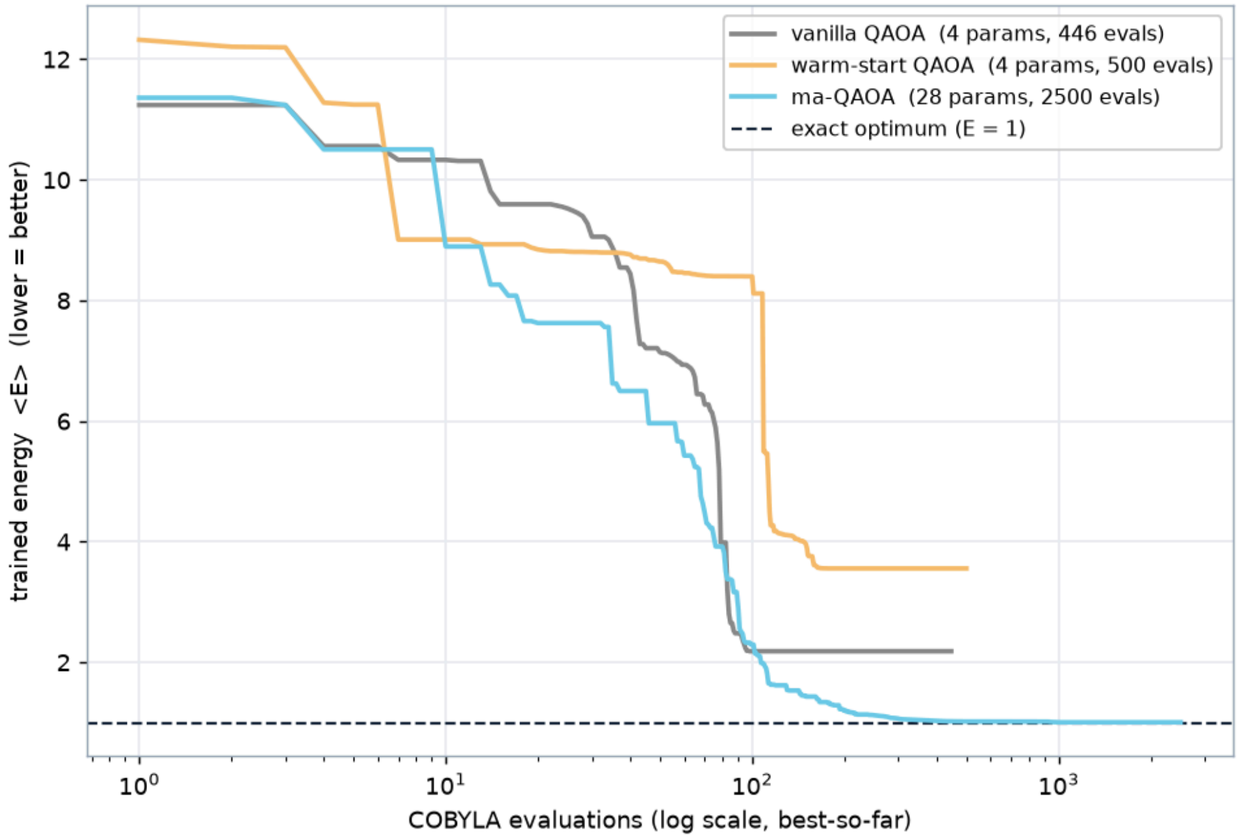
| variant | qubits | free params | two-qubit gates (CX) | two-qubit depth | logical depth | transpiled depth |
|---|---|---|---|---|---|---|
| vanilla QAOA | 5 | 4 | 36 | 24 | 15 | 49 |
| warm-start QAOA | 5 | 4 | 36 | 24 | 19 | 58 |
| ma-QAOA | 5 | 28 | 36 | 24 | 15 | 49 |

### Where the depth goes (and the two-qubit cost)

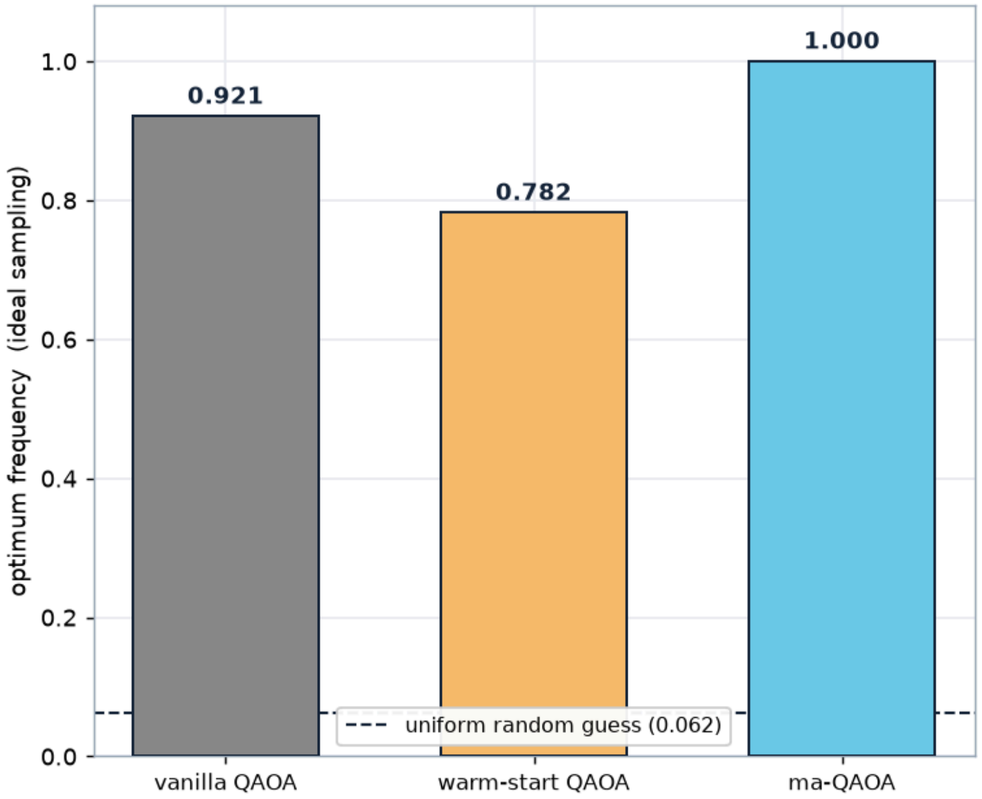


## QAOA ansatz comparison - convergence and accuracy (p = 2, five-SNP instance)

### Convergence of the classical optimiser

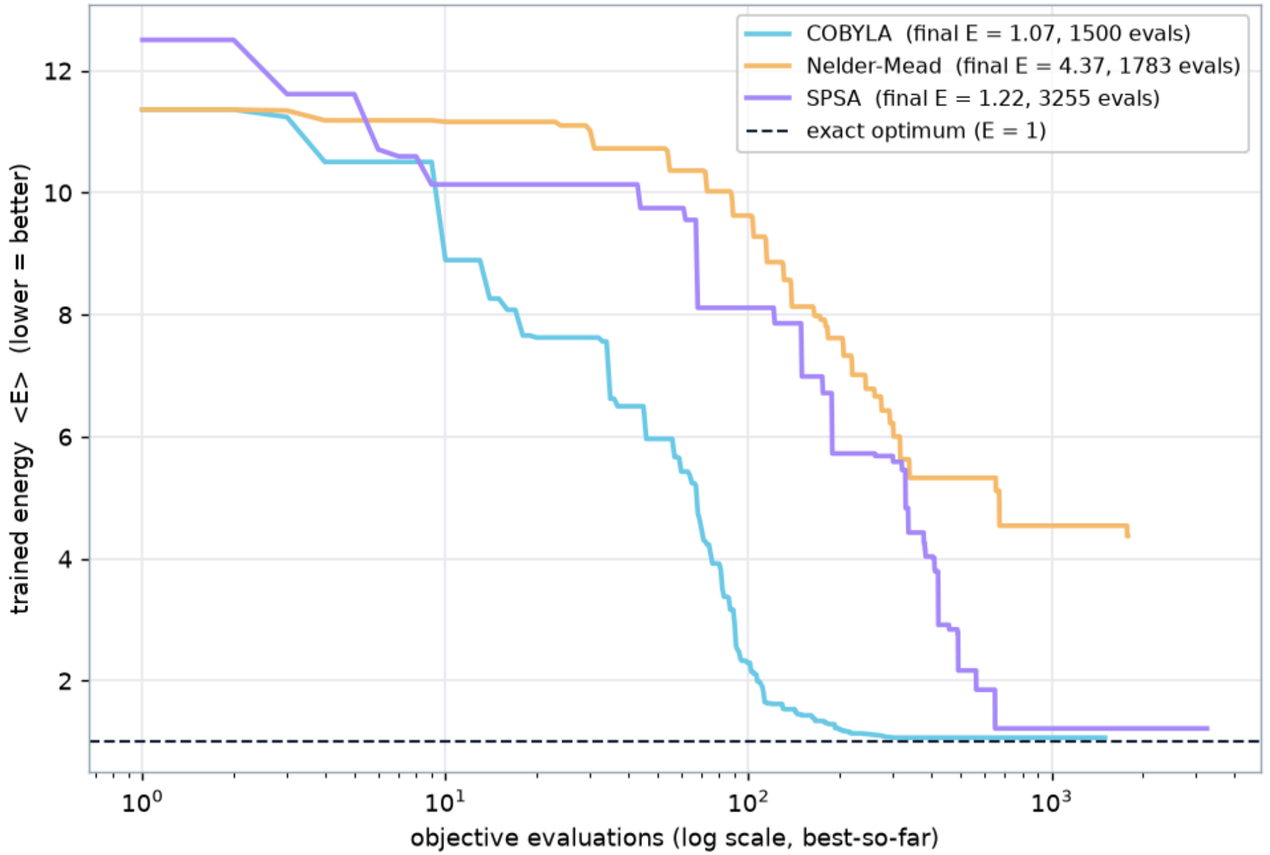



### Accuracy: how often the optimum is sampled

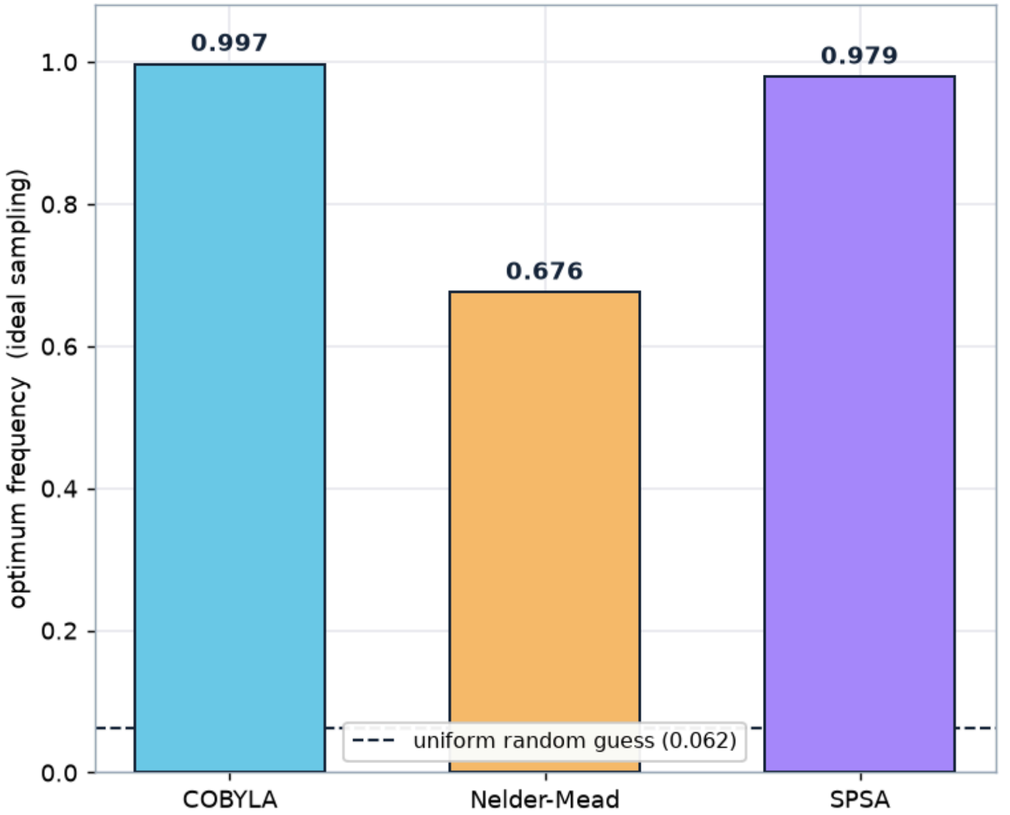


## ma-QAOA optimizer comparison - COBYLA vs Nelder-Mead vs SPSA (28 params, p = 2, five-SNP)

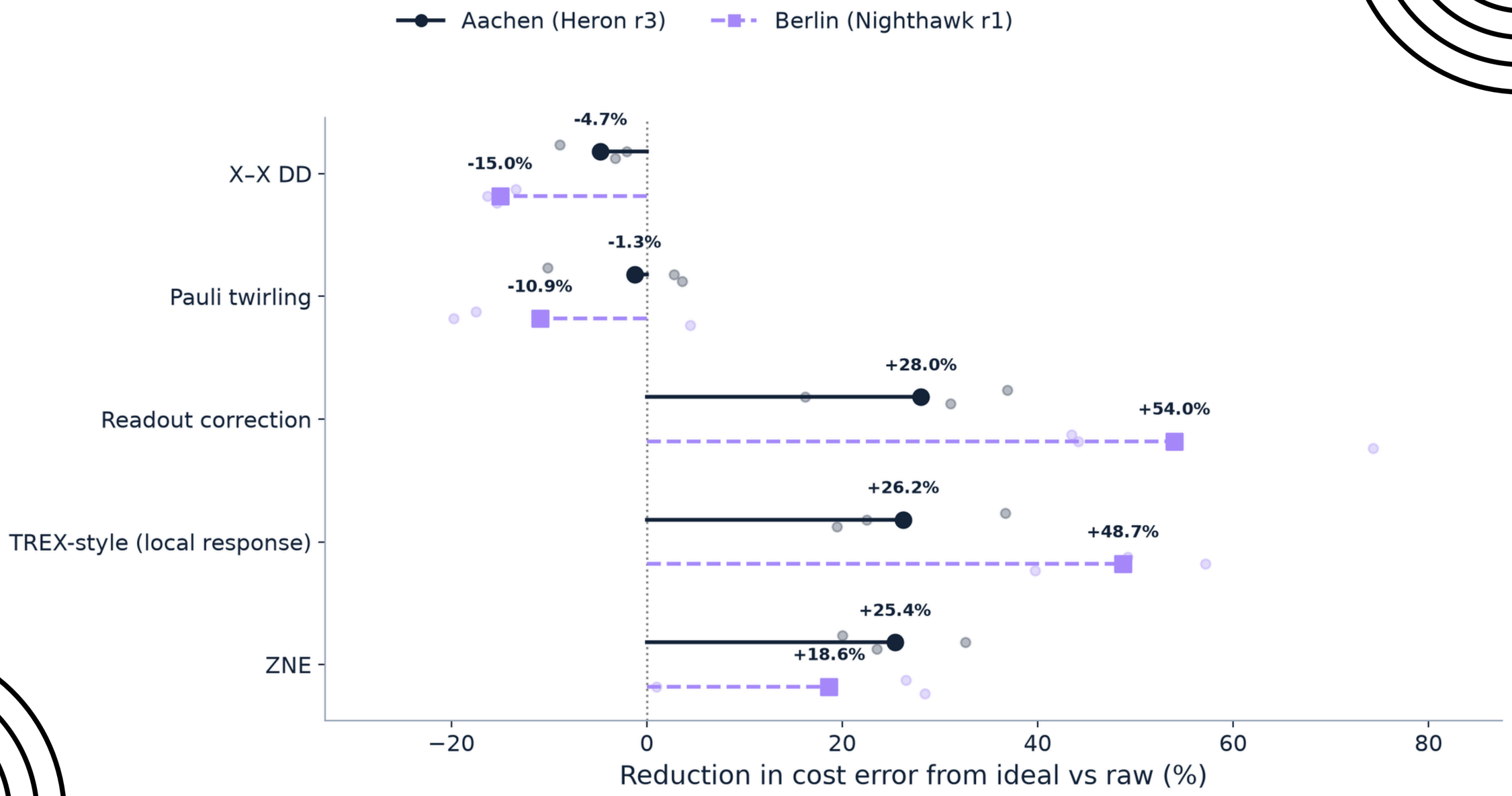
### Convergence of the classical optimiser



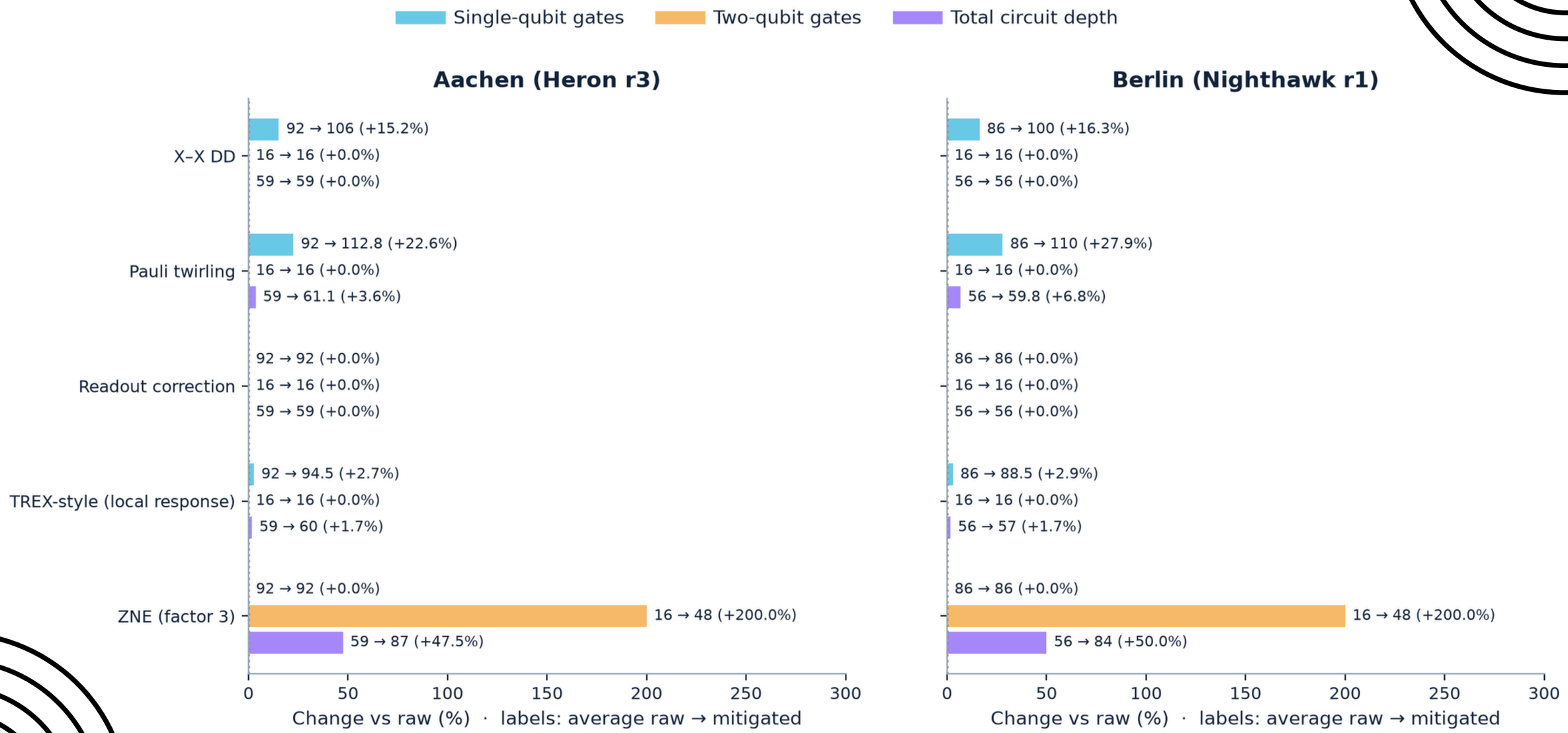

### Accuracy: how often the optimum is sampled

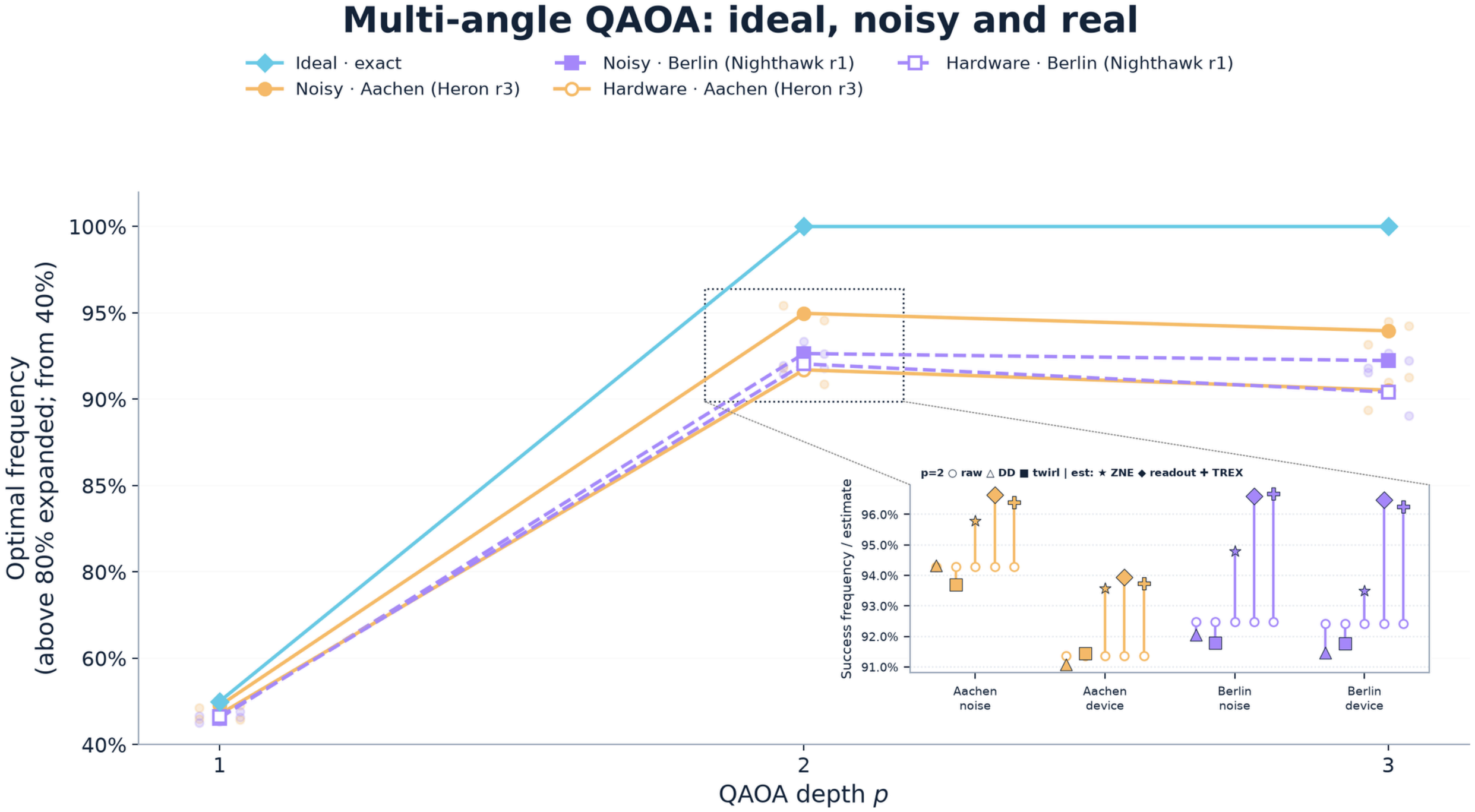


# Hardware: did mitigation help?

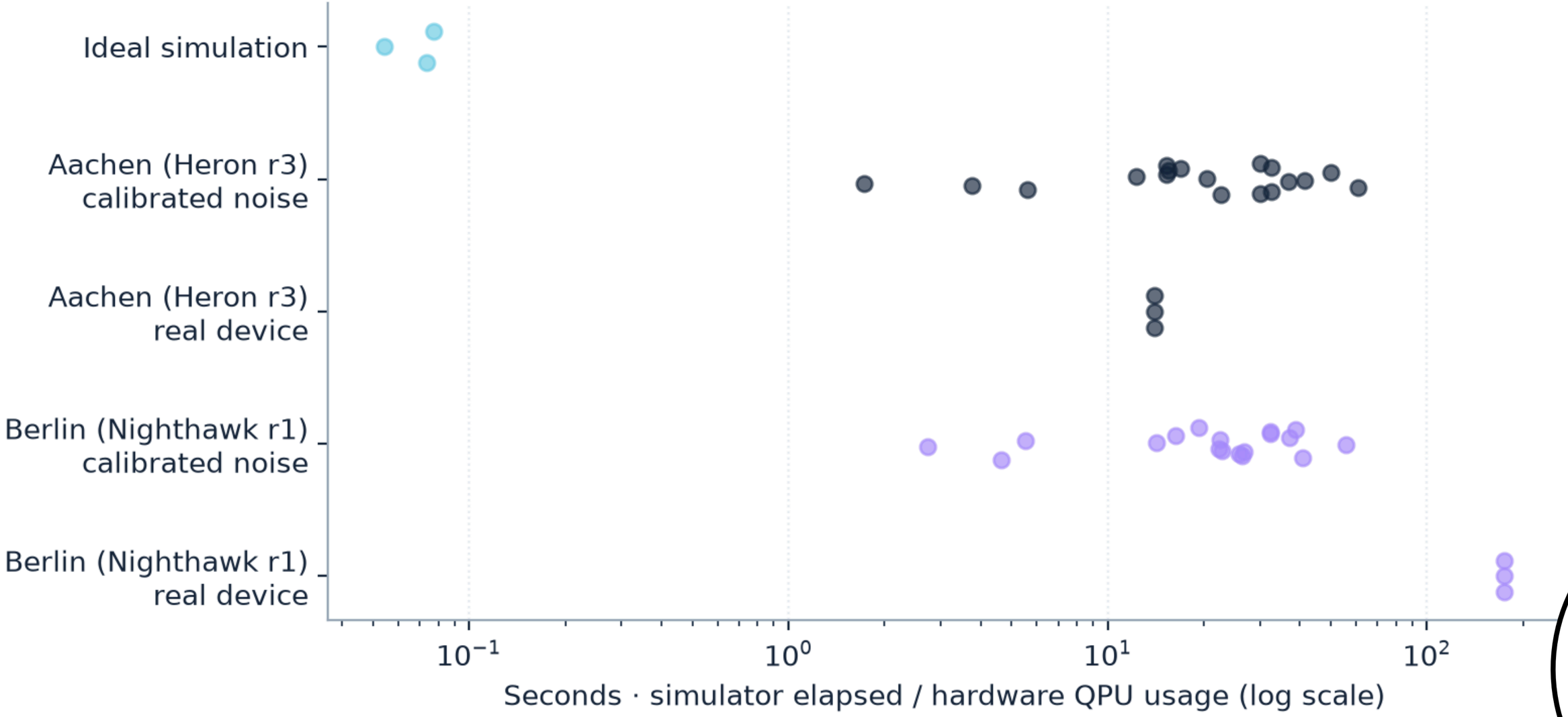


# Hardware: mitigation circuit resources




Multi-angle QAOA: ideal, noisy and real
Ideal · exact
Noisy · Berlin (Nighthawk r1)
Hardware · Berlin (Nighthawk r1)
Noisy · Aachen (Heron r3)
Hardware · Aachen (Heron r3)
Optimal frequency
(above 80% expanded; from 40%)
100%
95%
90%
85%
80%
60%
40%
1
2
3
QAOA depth p
p=2 ○ raw △ DD ■ twirl | est: ★ ZNE ◆ readout ✚ TREX
Success frequency / estimate
96.0%
95.0%
94.0%
93.0%
92.0%
91.0%
Aachen noise
Aachen device
Berlin noise
Berlin device

## Execution time by batch · excludes hardware queue



## Other hardware optimizations we tried

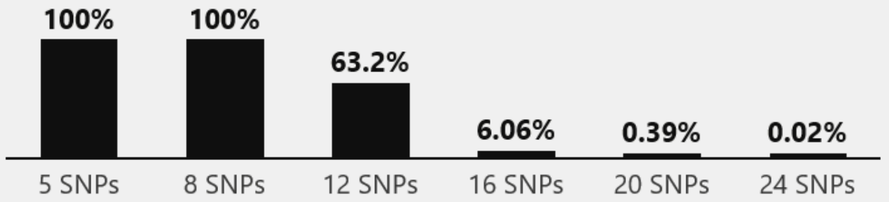
- SNP <-> Qubit Placement Order
- Gate Order →
  stronger interactions first
- Noise-aware layout →
  smaller errors, but higher depth

# Classical solvers vs QAOA

1. **Puzzles** — Synthetic reads — **8 → 2000** SNPs per puzzle
2. **Shared cost** — Read votes → J and offset — **1 cost** for every method
3. **Referee** — Brute force ≤ 24 SNPs, band dynamic programming beyond — **<0.15 s** exact optimum, 2000 SNPs
4. **Classical** — Random guessing · greedy + local search · annealing — **3 rivals** 64 independent runs each
5. **QAOA training** — p=1 - formula; p=2, 3 reuse angles (≈95% of tuned quality) — **0 shots** spent on QPU training
6. **Simulators** — Statevector ≤ 26 qubits, then MPS — **100 qubits** p=3 in 2.2 s
7. **Benchmark** — **2048 tries** per method

## Why the notebook's 5-SNP toy is too easy

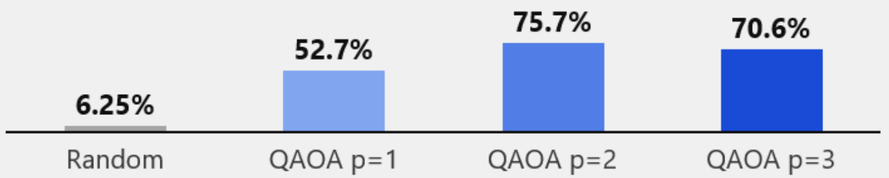
Chance that 2048 random tries hit the optimum




**5 SNPs = 32 answers → random guessing wins.**

## Zero-shot training works on the toy

Chance that one shot is optimal · exact simulation · 5 SNPs

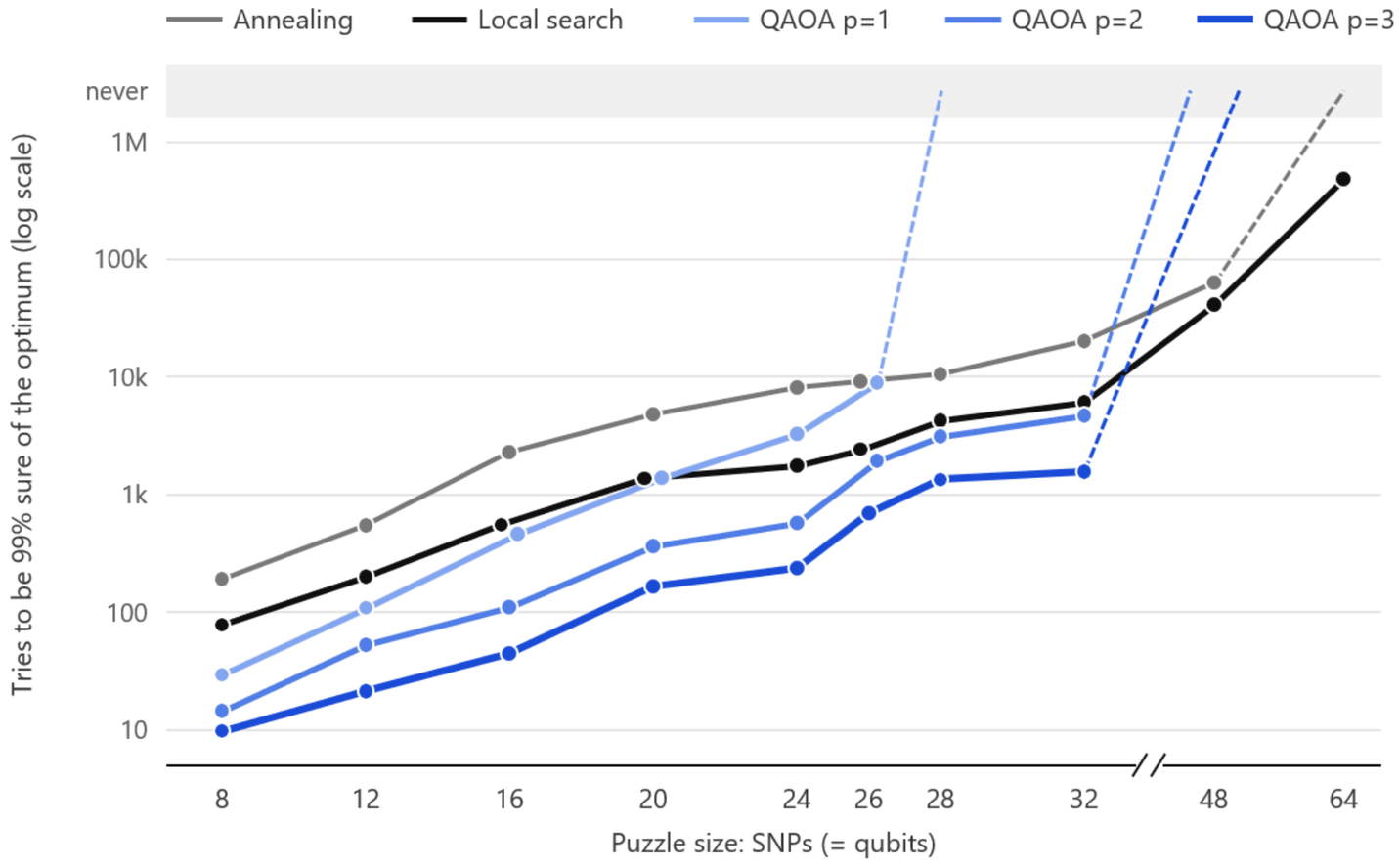



**p=1 formula:** cost 5.83 vs COBYLA's 7.80 (lower is better)

Puzzles throughout: coverage 6, reads spanning 2–4 SNPs, 5% read errors, 10% missing calls · QAOA angles: p=1 formula, layer-by-layer growth, reuse across puzzles

# Scaling up: QAOA leads on tries ≤ 32 SNPs, then stalls

Tries needed to be 99% sure of seeing the optimum (TTS99, median of 5 puzzles) · lower is better

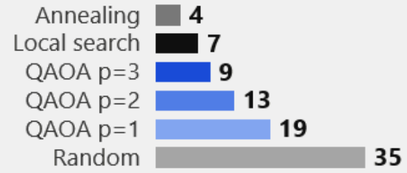



**How many times fewer tries QAOA p=3 needs than local search**

| 8 | 12 | 16 | 20 | 24 | 26 | 28 | 32 | 48 | 64 |
|---|---|---|---|---|---|---|---|---|---|
| **8.0** | **9.5** | **12.5** | **8.3** | **7.4** | **3.5** | **3.2** | **3.9** | **n/a** | **n/a** |

## At 100 SNPs: no optimal QAOA shot

Switch errors vs the hidden truth



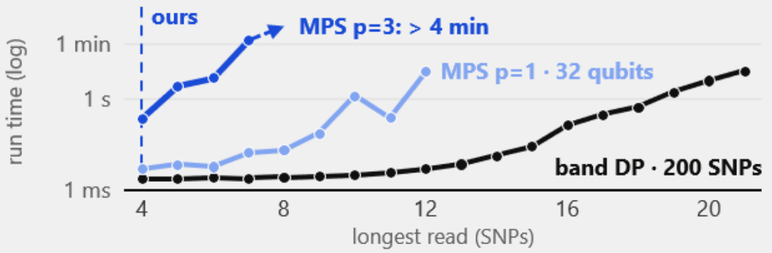

**Deeper QAOA helps:**
**19 → 13 → 9**
(p=1 → 3)

## Limit: classically easy

**SOLVE** Band DP: exact optimum of 2000 SNPs in 0.15 s.

**SIMULATE** MPS: 100 qubits in 2.2 s and 2 MB, not $2\times10^{31}$ B.

**What if reads were longer?**

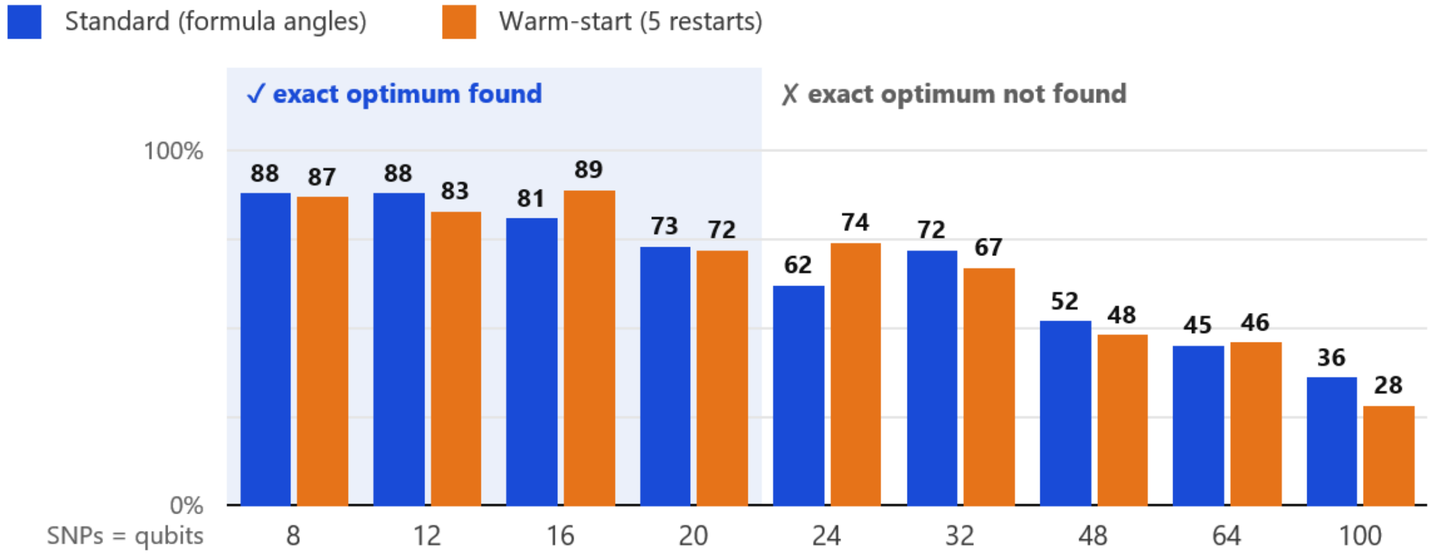



Our reads sit at the cheap end. Longer reads would break both shortcuts, but they also make the circuits deeper.

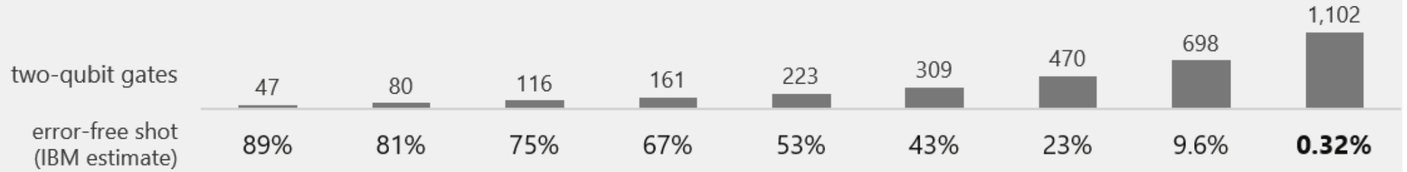
# Real QPU: exact answers up to 20 SNPs

IBM Aachen (Heron r3) · p=1 · 2,048 shots per circuit · one dataset per size · 8–100 SNPs

**Kept %: share of the noise-free gain that survives on the QPU**




**Why it fades:** bigger puzzles need more two-qubit gates, so fewer shots run error-free




**Better training: 0 → 18 optimal shots at 16 SNPs.** At 100 SNPs, 28–36% of the gain survives.

## ≤ 20 SNPs

**exact optimum found on the QPU**

1,048,576 possible answers
warm-start hit it 83× in 2,048 shots

**Best cost in 2,048 attempts (lower is better)**

| SNPs | Exact (DP) | QPU | Local search | Random |
|---|---|---|---|---|
| **16** | 7 | **7** | **7** | 14 |
| **24** | 13 | 14 | **13** | 27 |
| **48** | 24 | 46 | **31** | 91 |
| **100** | 57 | 146 | **75** | 192 |

### Takeaways

- **Fair race:** same puzzles, budget and exact referee
- **Simulation:** QAOA leads on tries ≤ 32 SNPs, then stalls
- **Classically easy:** DP solves it, MPS simulates it
- **Real QPU:** exact up to 20 SNPs; not an advantage claim

# THANK YOU

